



AMOD-5430H-A: Data Visualization 2026SU - Peterborough Campus

Instructor:

Instructor: Mikhail Genkin

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Office: OC 119

Office Hours: Monday, 2 p.m. to 4 p.m. in person, on-demand via Zoom

Meeting Times:

Please consult the Academic Timetable for the most up to date information.

[Graduate Calendar, Office of the Registrar](#)

Co-instructors and Teaching Assistants:

Please consult Blackboard for additional information.

Department:

Academic Coordinator: Krista Noonan

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Description:

Data visualization is a main step in the analysis of data in a wide range of scientific research areas as well as business applications. We will discuss general approaches and tools, and techniques for

the visualization of various types of data, including spatial data, graph data, and time series data.

Learning Outcomes:

By the end of the course, students should:

- be able to describe how data visualization fits into the data science lifecycle
 - be able to describe the relevant data types associated with various research areas
 - be able to describe the characteristics of the datasets associated with various research areas with respect to size and storage requirements
 - be familiar with visualization techniques and be able to discuss their advantages, disadvantages and computational requirements be familiar with various ways in which data can be visualized interactively on the web, and critically summarize these techniques be able to interactively visualize heterogeneous data in multiple ways
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Texts:

Required Text:

Show me the Numbers by Stephen Few, 2nd Edition

Readings:

Additional readings will be posted in Blackboard.

Assessments, Assignments and Tests:

Assignments:

There will be 2 assignments. The assignments will be weighted equally and consist of tasks to be solved and submitted individually. The assignments will be submitted electronically. Each assignment will focus on problems associated with the material presented in class, the assigned readings from the textbook, or any of the additional assigned readings during the weeks prior to the due date of the assignments. The length of the assignments may vary and each assignment will be assigned at least one week prior to the due date.

Workshops:

Students will participate in 4 workshops that will involve team discussions and/or hands-on exercises designed to reinforce lecture materials, and prepare the students for the assignments and the project. Workshops will be held either in-person during regularly scheduled class time, or on-line, depending on class size and progress. Students can work individually or in teams, and each student will individually submit their workshop deliverables in Blackboard. Attendance of workshops is mandatory. Students will not have the opportunity to repeat missed workshops.

Course project:

As part of this course the students will complete a course project. The students may work individually, or in small teams of up to 3 people. The course project will include the following deliverables: 1) the project proposal that describes what the students want to do, and how to do it; 2) the project paper describing the project final findings in IEEE conference format; 3) the project code, sample data, and a readme file that explains how to run their code. Students may choose to work in R or in Python. The project must demonstrate the use of data visualisation techniques to solve a data science problem. The students can choose their problem domain, and data set for the project.

Grading:

Assignments: 30% (15% each)

Workshops: 20% (4 workshops, worth 5% each)

Course project: 50%

Grade Total by Withdrawal Date:

Students will receive at least 25% of their grade by the withdrawal date of June 18.

Schedule:

The following schedule is tentative, and may be adjusted based on actual class progress.

Week of	Activities
Week 1 May 4	• Introductions • Course policies and grading • How to come up with an interesting project idea • The data science lifecycle and the role of visualization
Week 2 May 11	• Storyboarding
Week 3 May 25	• Please note that May 18th is a holiday - Victoria Day, and so there is no class • R Overview
Week 4 June 1	• Univariate Approaches • Assignment 1 due
Week 5 June 8	• Multivariate Approaches
Week 6 June 15	• Multivariate Data Continued • Project Proposal Due!
Week 7 June 22	• Colours • Assignment 2 Due!
Week 8 June 29	• How To Lie With Graphs
Week 9 July 6	• Text Data
Week 10 July 13	• Spatial Data
Week 11 July 20	• Infographics

Week 12
July 27

- Outlook, Special Topics
- Final project paper + code due!

Course Guidelines:

Some of the course activities may involve the use of on-line tools such as Blackboard, Zoom, Discord, and/or others. While most classes will be in-person, some classes may take place on Zoom due to circumstances such as inclement weather, or health advisories. When interacting with the Instructor and their peers via these on-line technologies students are expected to conduct themselves politely and professionally, and in-line with Trent University policies, and class policies that will be discussed during the introductory lecture. Failure to do so may result in the student losing access to these on-line tools, and the situation being referred to the university administration. Please also note that disrespectful conduct may result in the student being assigned a grade of 0 for their workshop grade (up to 20% of the overall course grade), as determined by the course Instructor.

Late assignment and project submissions will be penalised 10% per day, up to a maximum of 3 days, after which the student will receive a grade of 0 on their deliverable. Deadline extensions will not be allowed for any reason, except those involving serious circumstances, such as an illness or severe adverse family-related event. Should such circumstances occur, please reach out to the course Instructor right away to discuss appropriate accommodations. Circumstances such as being busy at work or with other courses, vacation, travel etc. will not be considered sufficient grounds for a deadline extension.

The use of Generative AI tools, such as ChatGPT, is not allowed, and may result in a grade of 0 being assigned.

Students are also expected to ensure access to sufficient computing resources and adequate network connectivity to be able to complete and deliver the presentations and the project, and access the required university systems such as Blackboard. Unless Trent University IT confirms a prolonged system outage on the university side, loss of network connectivity will also not be considered sufficient grounds for a deadline extension, so please don't leave your assignment and project submission to the last minute.

University Policies:

Grading Appeals Procedure

As described in the Graduate Calendar, students may appeal grades. Students should first attempt to resolve a matter informally with the initial decision-maker(s). Where informal resolution is not possible, a student has the right to make a formal appeal. The appeal process is described in the appeals procedure section of the Graduate Calendar.

Equity

The School of Graduate Studies upholds Trent University's policy on rights and responsibilities of all members of the University states as follows: "Every member of Trent University – faculty, staff or student – has a right to freedom from discrimination in the University by another faculty, staff or student member because of race, ancestry, place of origin, colour, ethnic origin, citizenship, creed, sex, sexual orientation, age, record of offences, marital status, family status or handicap." (For details, please consult Trent's Policy on Discrimination and Harassment on the Trent University Centre for Human Rights, Equity & Accessibility web page.)

Access to Instruction

It is Trent University's intent to create an inclusive learning environment. If a student has a disability and/or health consideration and feels that he/she may need accommodations to succeed in this course, the student should contact the Student Accessibility Services Office (SAS), (BH Suite 132, 705-748-1281 sas@trentu.ca). More information can be found on the [Student Accessibility Services](#) website.

Graduate Academic Integrity Policy

Academic dishonesty, which includes plagiarism and cheating, is an extremely serious academic offence and carries penalties varying from failure on an assignment to expulsion from the University. Definitions, penalties, and procedures for dealing with plagiarism and cheating are set out in Trent University's *Academic Integrity Policy*. You have a responsibility to educate yourself – unfamiliarity with the policy is not an excuse. You are strongly encouraged to visit Trent's Academic Integrity website to learn more: www.trentu.ca/academicintegrity.

All members of the University community share the responsibility for the academic standards and reputation of the University. When students submit work for academic evaluation and credit, they imply that they are the sole authors of the work. Clear and careful attribution of the words and ideas of others is an essential part of academic scholarship. Academic honesty is a cornerstone of the development and acquisition of knowledge and is a condition of continued membership in the University community.

Student Absenteeism, Missed Tests and Examinations

Students are responsible for completing all course requirements, including attending classes and meeting assignment deadlines as specified on their syllabus.

Adjustments and deferrals to dates for participation, assignment submissions, tests, midterms and final examinations are not automatic. It is the student's responsibility to email their instructor immediately if they are unable to fulfill academic requirements.

Courses delivered remotely may involve student participation in scheduled (synchronous) classes via web-based platforms, such as Zoom. Students unable to participate (i.e., by video and/or audio) should email their instructors to request alternative arrangements for participation in these scheduled (synchronous) classes.

Students are required to be available for all tests, midterms and exams that are listed in their course syllabus and scheduled by their instructor or the Office of the Registrar. Depending on

their program, the instructor or the chair/director may decide on alternative arrangements for exams and tests. Normally a doctor's note or supporting documentation is not required; however, when a student's success in the course or program is in jeopardy as determined by the instructor or chair/director, documentation may be requested.

Specific SAS accommodations can be implemented for students registered with Student Accessibility Services (SAS), but it is the responsibility of the student to make these arrangements in advance as per SAS guidelines, and to discuss accommodations of due dates with their instructors.

Students can notify the Office of the Registrar of their wish to observe cultural or religious holidays during scheduled examination periods by the deadline set in the Academic Calendar. Personal travel plans are not acceptable reasons for missing tests or exams.

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